3. Run

3.1 Start the GUI

There are two copies of TreeSync, for two jobs:

	The installed TreeSync	**The build in the source tree**
Start it	**TreeSync** in the desktop's app list, or `treesync`	`v001/out/gcc/treesync` (or `out/clang/`)
Syncs as	**root** for an administrator (each run through `pkexec`); anyone else: themselves	you; root only with **Sync > Run as root**
Paths	any (administrator); inside their home (anyone else)	only inside your home, unless Run as root
Build page	none: it only runs	yes (File > Build..., and at start if ticked)
Comes from	`./install.sh` (see 2.6)	`./build.sh` or `./verify.sh`

For everyday syncs use the **installed** TreeSync. The source-tree build is for working on TreeSync itself: it starts on the Build page if **Show this page when TreeSync starts** is ticked (click **Continue to TreeSync using GCC exe** to go on).

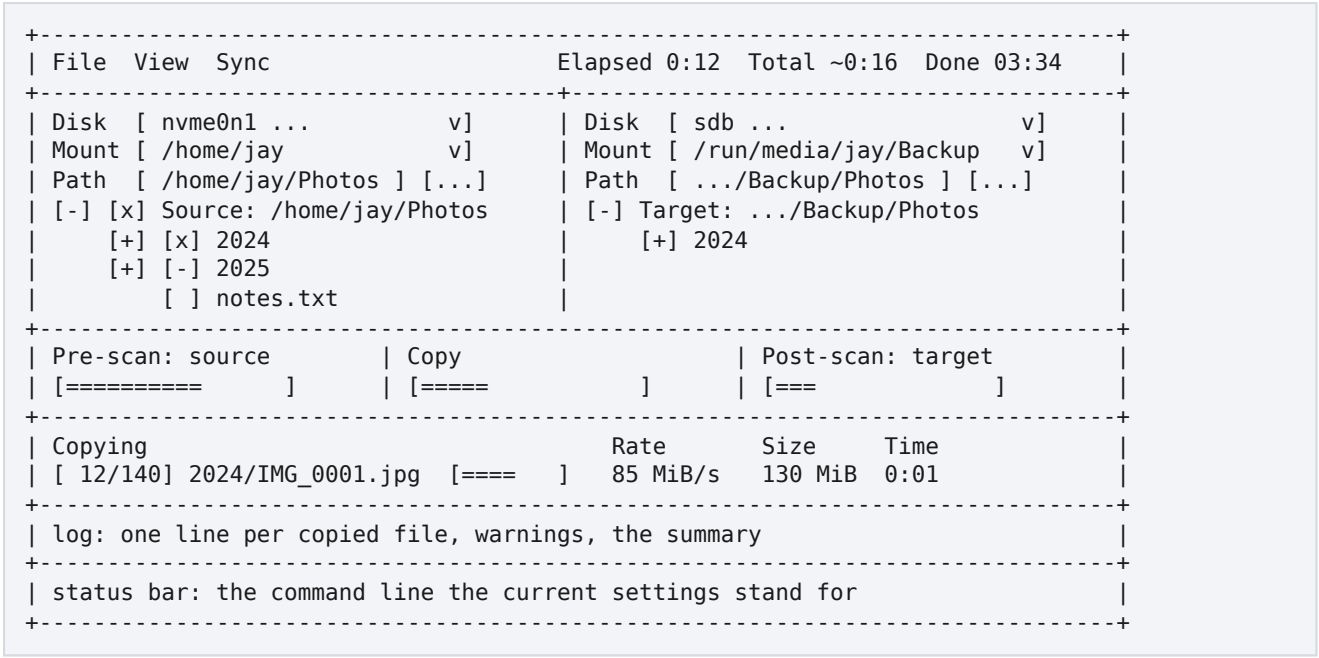
Always start the GUI as yourself, not with `sudo`: it is an ordinary desktop program, and only the syncs run as root.

Options on the command line set up the window and become the saved settings:

```
treesync --source=/home/jay/Photos --target=/run/media/jay/Backup/Photos --mirror
```

Without options, the settings saved last time are used (`~/.config/treesync/settings`, saved on every change).

3.2 The main window



- **Left: the source**, **right: the target**. Each side has a **Disk** and **Mount** list to jump to a drive, a **Path** box (type a path and press Enter, or **...** to browse), and the tree below it.
- **Trees**: click [+] to open a folder, or use the + and - keys. **View > Reload trees** reads them again, **View > Collapse all** closes them.
- **The progress boxes** and the list of files in progress appear while a run is going. The upper right of the menu bar shows how long it has run, how long the whole run will probably take, and when it should be done.
- **The log** shows what happened; **View > Show log file** (Ctrl+L) opens the full log of the newest run.

Without root, local paths are limited to your home folder; the status bar says so.

Help opens this manual: the whole of it, or one chapter, as a PDF. **F1** opens the chapter for the window you are in.

3.3 Choose what to sync: the checkboxes

Every folder and file in the **source** tree has a checkbox.

- **Check a folder**: everything below it is synced, including files added there later.
- **Uncheck a folder**: nothing below it is synced.
- **Uncheck one file** inside a checked folder: the rest of that folder stays checked.
- A folder with only some of its contents checked shows a **dash** (partly checked).
- The **top box** checks or unchecks the whole tree.

What is checked is saved as `--include-path` settings. Folders above a checked item are created on the target only where something checked goes, and **Mirror never deletes anything outside the checked paths**. Changing the source folder starts again with everything checked. If nothing is checked, Run refuses to start.

3.4 Setup: mode and options

File > Setup... has three tabs.

Source tree / Target tree: the path, and for a remote machine the **Remote** box with **Host**, **Port**, **User** and **Key file** (see 3.10).

Flags:

- **Mode**:
  - **Copy**: new and changed files go from source to target; nothing on the target is removed. Turns on *Skip older*: a newer file on the target is not overwritten.
  - **Mirror**: the target is made to match the source; files only the target has are removed (with safe delete: moved to `<target>.old`).
  - **Merge**: changes go both ways; a file changed on both sides since the last sync is a conflict and is left alone. Nothing is removed.
- **Reset to the defaults** of Copy, Mirror or Merge.
- **Owners**: don't copy / by name (normal) / by number. Setting an owner needs root.
- **Switches**: click any option to see what it does in the help box. The important ones: *Safe delete* (on; keep it on), *Hard links*, *Follow links*, *Skip older / newer / changed*, *Keep extra*, *Wipe target first*, *Low priority*.
- **Numbers**: threads (default 4), retries, chunk size, bandwidth limit, how long to wait for a lost network, time tolerance.
- **Include / exclude**: file and folder names with `*` and `?`, separated by `;`. A blank box filters nothing; exclude wins over include.

3.5 Dry run, then run

1. **Sync > Dry run** (Shift+F9): lists what a run would do (`copy`, `link`, `delete`, `attrs`, `mkdir`) and changes nothing. Read it.
2. **Sync > Run** (F9): asks for confirmation, saying what the mode will do and where removed and replaced files go, then runs.

While it runs:

- **Pre-scan: source** walks the source. **Copy** shows the bytes copied of the total. **Post-scan: target** checks each target folder for files to remove (Mirror).
- Each file is written to `NAME.treesync-part`, read back, checked against the source's SHA-256, given the source's permissions, times, owner and extended attributes, and only then renamed into place.
- A file whose content is the same but whose permissions or time changed gets only its attributes set (an `attrs` step), without copying.

At the end a popup says how it went and how long it took. **Sync > Stop** (Esc) stops after the current file or chunk.

3.6 Stop and resume

A stopped run (Stop, Ctrl+C in a terminal, a crash, a reboot, an unplugged disk) is **resumed by running it again with the same settings**. Every step was recorded in the database, so it continues where it was. A big file continues at its last good chunk: on resume, every chunk already copied is read back and checked first, and from the first damaged one the file is copied again.

To start over instead, use `--fresh` on the command line.

3.7 Checking and attributes

- **Sync > Scan only (SHA-256 check)** (F7): changes nothing. Lists both trees, reads and hashes **every file on both sides** afresh, and compares permissions, times, owners and extended attributes. Each file's log line ends in `same`, `DIFFERS`, `missing` (not on the target) or `failed`. Use it after a big copy to prove the target is right.
- **Sync > Sync attributes** (F8): sets only the attributes that differ (mode, time, owner, extended attributes) on the target, for files of the same size and for folders. No content is copied.
- **Sync > Clear .old folders...**: deletes the safe-delete folders (`<target>.old`) for good, after a confirmation that says how many files and bytes. Folders made by a root run need **Run as root**. On the command line, without `--no-dry-run` it only counts; its result says *Counted, not deleted*, *Good* (deleted) or *Nothing to clear*.

3.8 History, logs and AI export

- **View > History...** (Ctrl+H): every run, newest first, with its counts (folders, copied, deleted, attributes set, failed, conflicts, bytes). Failed runs are red. Below:
  - **Actions**: what the selected run did, failures first.
  - **SHA-256 checks**: every checked file under a line naming its folder, with the source and target SHA-256 side by side, whether they match, and how the attributes compare. Mismatches first, in red; attribute differences in orange.
  - **Show log**, **Export logs to AI.md**, **Export SQL to AI.md**.
- **Logs**: each run writes `logs/YYYY-MM-DD_HHMMSS.log` next to the database. It starts with a header; every line has its start, stop and duration to the microsecond.
- **View > Export newest run for AI**: writes the run as Markdown files into `logs/ai/` (a guide, problems, slowest copies, throughput, the logs in full; or the database contents), ready to give to an AI to analyse.

3.9 Running as root

A sync often meets files your user cannot read or replace (other users' files, system files), so the installed TreeSync **syncs as root for an administrator**: root, or a member of `wheel` (`sudo` or `admin` on Debian, Ubuntu and Mint), the users who may use sudo. A tool that can overwrite any file as root is as powerful as root, so nobody else gets that: **anyone else runs the installed TreeSync as themselves**, inside their own home, with no password asked (the Sync menu shows *Run as root (administrators only)*, greyed out).

For an administrator:

- **The installed GUI** runs every sync through `pkexec`. The first run asks for your password; polkit then remembers it for a few minutes. **Sync > Runs as root (installed)** is shown ticked and cannot be turned off. Any path may be chosen.
- **The installed `treesync-cli`**, started from a terminal by an administrator, starts itself again through `sudo`: `treesync-cli --source=... --target=...` runs as root (sudo asks for your password). `--help` does not need it.
- **The source-tree build** syncs as you, limited to your home folder. Ticking **Sync > Run as root (pkexec)** makes its runs use the **installed** `treesync-cli`, which only root can change. If that is not this build, the log says so (`File > Install for root` updates it); if nothing is installed, it warns that it runs a copy your user could change.

A root run keeps its own database and logs under `/root/.local/share/treesync/`, separate from yours. To resume a root run, run it as root again.

History and logs of root runs. The GUI's **History** and **Show log file** read *your* database and logs, and root's are not readable by you. To look at a root run: `sudo ls /root/.local/share/treesync/logs/`, or have root export it: `treesync-cli --export-ai-logs` (the installed one runs as root by itself; it writes into root's `logs/ai/`). Note that `sudo treesync-cli` does not work on RHEL: sudo does not search `/usr/local/bin`; the installed `treesync-cli` needs no sudo in front.

3.10 Remote trees (SFTP)

Either side can be on another machine over SSH. One-time setup:

```
ssh-keygen -t ed25519            # if you have no key yet
ssh-copy-id jay@server           # passwordless login
ssh jay@server                   # once, to check and accept the host key
```

TreeSync logs in with keys only, never passwords. In the GUI: **File > Setup > Target tree**, tick **Remote**, and fill in Host, Port, User (and Key file if not the usual one). On the command line: `--target-host=server --target-user=jay`.

If the network drops, the run waits (default up to 30 minutes, *Wait for the network*), shows **NETWORK UNAVAILABLE**, and carries on when the machine is back.

Checking copies on the server. When the login may run commands and the server has `sha256sum`, the server hashes its own files: copies are checked there and remote scans hash there, so nothing is read back over the network. The log says which way it works (`server hashes its own files` or `files are read over SFTP`). An SFTP-only account still works, just with twice the traffic for a verified copy.

3.11 The command line

`treesync-cli` takes the same options the GUI saves. **It does a dry run unless you give `--no-dry-run`.** The examples use the installed `treesync-cli` (it runs as root); in the source tree it is `v001/out/gcc/treesync-cli` (it runs as you).

```
# What would a mirror do?
treesync-cli --source=/home/jay/Photos --target=/run/media/jay/Backup/Photos --mirror

# Do it
treesync-cli --source=/home/jay/Photos --target=/run/media/jay/Backup/Photos --mirror --no-
dry-run

# Copy only new and changed files, 8 threads, at low priority
treesync-cli --source=/data --target=/backup/data --copy --threads=8 --low-priority --no-dry-
run

# Only one folder and one file of the source
treesync-cli --source=/data --target=/backup/data --include-path=projects/2026 --include-
path=notes.txt --no-dry-run

# To a remote machine
treesync-cli --source=/data --target-host=server --target-user=jay --target=/backup/data --
no-dry-run

# Prove a copy: hash both sides, change nothing
treesync-cli --source=/data --target=/backup/data --scan-only

# Set only differing permissions and times
treesync-cli --source=/data --target=/backup/data --sync-attrs --no-dry-run

# Delete <target>.old for good (without --no-dry-run it only counts)
treesync-cli --source=/data --target=/backup/data --clear-old --no-dry-run
```

`treesync-cli --help` lists every option. In a terminal it shows one line per copied file, `[ 6/90] name 100% | 85.3 MiB/s | 130.9 MiB | 00m01s`, and Ctrl+C stops it (press twice to quit at once).

Exit codes: 0 in sync (or, for Scan only, the trees are the same); 1 failures, conflicts or differences; 2 could not run (bad options, a path refused); 3 stopped.

Database views for looking at results directly:

```
sqlite3 ~/.local/share/treesync/treesync.db
```

```
SELECT run, started, state, copied, failed FROM run_summary ORDER BY run DESC LIMIT 10;
SELECT * FROM failures WHERE run = 42;
SELECT path, file_name, problem FROM hash_mismatches WHERE run = 42;
SELECT path, file_name, attr_diff, result FROM attr_mismatches WHERE run = 42;
```